1. Introduction

The time series continuation of financial asset returns[1] has been widely studied in the academic literature, through investigating serial correlation (Fama and French, 1988; Lo and MacKinlay, 1988; Lewellen, 2002; DeMiguel et al., 2014) and time series momentum (TSM) (Moskowitz et al., 2012; Menkhoff et al., 2012; Georgopoulou and Wang, 2016). One practical application of time series continuation is the trend strategy which in recent years has become increasingly popular among hedge funds. However, what has not yet been examined exhaustively is the reversal effect which emerges after the continuation pattern.[2]

This paper empirically investigates the reversal property of various financial assets and analyses its relationship with continuation focusing exclusively on the time series dimension. Return continuation and reversals are usually closely related and discussed together in the literature. Historically, most studies have focused on cross-sectional momentum and reversals which have been documented internationally and in various financial assets.[3] According to a number of well-known behavioural theories by Barberis et al. (1998), Daniel et al. (1998), and Hong and Stein (1999) among others, the rationale behind momentum and reversals is related to short-term under-reaction and delayed over-reaction. In the time series context, Moskowitz et al. (2012) also attribute the TSM effect to these behavioural features. Some recent studies attempt to explain the momentum and reversal effect using rational expectation models, for example, information percolation (Andrei and Cujean, 2017), heterogeneous beliefs (Ottaviani and Sørensen, 2015) and time-varying factor loading (Kelly

[1]Return continuation is seen in the finance literature as an analogue of momentum; see, among others, Rouwenhorst (1998) and Fama (1998). Sometimes these two words are used interchangeably. However, return continuation is applicable across a wider range of dynamics.

[2]Though some studies in the literature mention it, for example, Moskowitz et al. (2012) uncovered a long-term reversal beyond one year of TSM signals.

[3]Cross-sectional momentum and reversals have been extensively studied in various financial markets, see, e.g., the US stock market (De Bondt and Thaler, 1985; Lo and MacKinlay, 1990), international stock markets (Fama and French, 1998; Rouwenhorst, 1998), country indices (Bhojraj and Swaminathan, 2006), various asset classes (Asness et al., 2013), and commodities (Miffre and Rallis, 2007; Gorton et al., 2013).